

Rajeev Marada

✉ rajeevmarada02@gmail.com  rajeevmarada  RajeevMarada  Engineering Portfolio

PROFESSIONAL SUMMARY

Dedicated VLSI engineer with advanced certification from IISc Bangalore and hands-on experience in digital design, verification, and UVM methodology. Proven ability to translate complex theoretical concepts into efficient, real-world hardware solutions, demonstrated through published research and a granted patent. Passionate about developing reliable, high-performance hardware for AI and next-generation computing challenges.

EDUCATION

- PG Level Advanced Certification in VLSI Chip Design, Indian Institute of Science (IISc)** Feb 2024 – Dec 2024 | Bangalore, IN
- Completed an intensive 11-month program, achieving an 80% overall score.
 - Relevant Coursework: Analog & Digital Integrated Circuit Design, Advanced Digital Design & FPGA-based Design, RISC-V Architecture.
- Bachelor of Technology in Electronics and Communication Engineering, SRM Institute of Science and Technology** Jun 2019 – May 2023 | Chennai, IN
- Graduated with a CGPA of 9.4/10.0.
 - Relevant Coursework: Digital Design, ARM-based Embedded System Design, VLSI Design Methodology.

PROFESSIONAL EXPERIENCE

- Programmer Analyst, Cognizant Technology Solutions** Aug 2024 – Present | Chennai, IN
- Automated key business workflows within the Intelligent Process Management team by developing and deploying solutions on the APPIAN low-code platform.
 - Collaborated on cross-functional projects to analyze and optimize enterprise-level processes, contributing to enhanced business efficiency.
- Junior Design Verification Engineer, Insemi Technology Services** Jul 2023 – Oct 2023 | Bangalore, IN
- Developed a comprehensive UVM testbench to validate a Dual-Port RAM design, creating constrained-random test cases that achieved 100% functional coverage.
 - Analyzed protocol compliance by writing SystemVerilog assertions and functional coverage, successfully identifying and debugging design flaws.
- Intern, Maven Silicon** Dec 2022 – Jan 2023 | Bangalore, IN
- Deepened understanding of digital and system-level design fundamentals, especially finite state machine (FSM) architecture, and enhanced practical skills in module integration and testbench planning.
 - Designed and implemented a compliant AMBA AHB-APB bridge in Verilog to manage communication between high- and low-frequency SoC subsystems.
- Intern, CoreEI Technologies – Sandeepani School of Embedded System Design** Jun 2022 – Jul 2022 | Bangalore, IN
- Learned SystemVerilog verification workflows, building key testbench components—drivers, monitors, scoreboards, assertions—and applied digital design principles for module development.
 - Developed and validated a SystemVerilog testbench for a full adder DUT, achieving 100% functional and code coverage through rigorous assertion-based verification.

KEY PROJECTS

- RISC-V Based Hardware Accelerator for ECG Classification** Feb 2025 – Apr 2025
- Engineered a heterogeneous SoC integrating a RISC-V processor and a custom neural network via the AXI Stream protocol, achieving 92% accuracy in real-time ECG classification through Q1.15 fixed-point arithmetic optimization.
 - Validated the complete SoC and accelerator design through simulation and deployment on FPGA hardware, demonstrating robust real-time performance and ensuring practical hardware deployability.
- Neural Network Inference Engine for Handwritten Digit Classification** May 2024 – Dec 2024
- Designed and implemented a Verilog RTL 3-layer neural network inference engine with custom ROM-based weights and FIFO input, achieving over 85% Semeion dataset accuracy through efficient fixed-point hardware conversion.
 - Performed simulation, functional verification, synthesis, and implementation in Xilinx Vivado, establishing timing closure and implementation readiness for FPGA-based deployment.
- UVM Testbench for APB Protocol Verification** Dec 2023 – Jan 2024
- Architected a modular UVM testbench, developing reusable master/slave agents and environment components, to rigorously validate APB protocol compliance.
 - Achieved 100% functional and code coverage and confirmed design robustness through complete assertion coverage, ensuring error-free operation.
- Error Correction Module for SoC Data Integrity** Jan 2023 – May 2023
- Led development of a novel SEC-DED-DAEC error-correcting module and integrated it into an AHB-APB bridge to ensure data integrity for system-on-chip pathways.
 - Optimized the reversible logic implementation to achieve a 5.67% reduction in power consumption and a 4.52% improvement in critical path delay over other novel designs, culminating in an IEEE publication.

PATENT

- A System for Controlling an Autonomous Vehicle and a Method Thereof** Mar 2025
- Patent Number:** IN 564400 | **Status:** Granted
 - Addressed the critical safety risk of driver incapacitation by developing a real-time controller for vital-sign sensors and designing an actuator feedback loop, resulting in a successful prototype that could automatically detect medical distress and initiate a safe autonomous takeover.

PUBLICATIONS

Improving data integrity with reversible logic-based error detection and correction module on AHB-APB bridge, <i>Conference Paper - IEEE</i>	May 2023
A Survey on Affordable Internet of Things (IoT) Enabled Healthcare Systems, <i>Journal Article - Grenze International Journal of Engineering & Technology (GIJET)</i>	Jun 2022

TECHNICAL CONFERENCES

IEEE International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence, <i>(RAEEUCCI 2023)</i>	Apr 2023
Twelfth International Conference on Advances in Computer Engineering, <i>(ACE 2022)</i>	May 2022

CERTIFICATIONS & ONLINE COURSES

SystemVerilog for Verification Part 1: Fundamentals, <i>Udemy</i> Acquired foundational knowledge of SystemVerilog and its application in digital verification workflows.	Apr 2023
Verification Methodology Overview, <i>Maven Silicon</i> Gained a comprehensive overview of hardware verification strategies, focusing on real-world DUT testing approaches.	Nov 2022
VLSI Design Methodologies, <i>Maven Silicon</i> Developed a thorough understanding of digital design, Verilog coding, and core concepts in modern VLSI architectures.	Oct 2022
Development of Real-Time Systems, <i>Coursera/EIT Digital</i> Explored the design and importance of real-time systems for reliable, time-critical embedded applications.	Mar 2022
Arduino Training, <i>Internshala</i> Gained a foundation in electronics concepts and implemented Arduino circuits for IoT and automation projects.	Feb 2022
PCB Design Training, <i>Internshala</i> Designed foundational PCB layouts using Autodesk Eagle, applying principles of circuit design, via placement, and multi-layer board construction.	Nov 2021
Blockchain Basics, <i>Coursera/University at Buffalo, SUNY</i> Studied core mechanisms and practical applications of blockchain technology, cryptocurrency, and decentralized systems.	Aug 2021

WORKSHOPS & TECHNICAL TRAINING

Kaizen IoT Workshop, <i>Lema Labs</i> Designed and implemented a home automation system using MQTT protocol for IoT device communication.	Mar 2022
Intel Distribution of OpenVINO Toolkit Online Training, <i>Intel</i> Gained exposure to deployment practices for AI, machine learning, and computer vision models using Intel’s OpenVINO toolkit through hands-on demonstrations and expert-led instruction.	Mar 2021

TECHNICAL SKILLS

Domains: Digital Design & Verification SoC Architecture Neuromorphic Computing
Languages: SystemVerilog Verilog Python C C++ LaTeX
Methodologies: Universal Verification Methodology (UVM)
Tools & EDA: Xilinx Vivado ModelSim Questa Sim Cadence Virtuoso MATLAB

HONORS & AWARDS

Finalist, Hybrid Hack 2021 ("Charge On Go") - Co-developed a novel vehicle-mounted hybrid power system to generate electricity during transit and supply it back to the grid. - Simulated the system's electrical performance using Python and HOMER Pro, projecting an annual energy generation of 465.15 kWh per unit based on an idealized motion dataset.	Sep 2021
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VOLUNTEER EXPERIENCE

National Service Scheme (NSS) Volunteer - Collaborated with NSS team members and contributed ideas for community outreach and improvement projects. - Helped organize and lead medical camps focused on care and health awareness for underserved groups. - Assisted in coordinating anti-drug awareness campaigns and local educational events for youth and families. - Took initiative in environmental activities, including water conservation, tree plantation, and cleanliness drives.	Jun 2019 – Mar 2020
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SOFT SKILLS

Resilience and Perseverance Adaptability Critical Thinking Teamwork and Collaboration

INTERESTS

Photography Music eSports Cricket
